



BHARATHIYAR ARTS AND SCIENCE COLLEGE FOR WOMEN

Department of Computer Science with Cyber Security

</> PROJECT TITLE

AI STUDYBUDDY API

An Intelligent Backend for AI-Powered Learning with Summaries, Flashcards, Quizzes and Personalized Study Plans.



PROJECT DEVELOPMENT TEAM



TEAM LEADER

A.NAVEENA

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LearnMate — AI Assistant for Smarter Learning

AI-powered learning platform | Technical Project Presentation

AI STUDYBUDDY

Generate • Revise • Practice • Plan

React.js

Node.js + Express

MongoDB + Mongoose

Google Gemini API

JWT + RBAC

1. Problem & Proposed Solution

From scattered study materials to one intelligent learning workflow

Problem

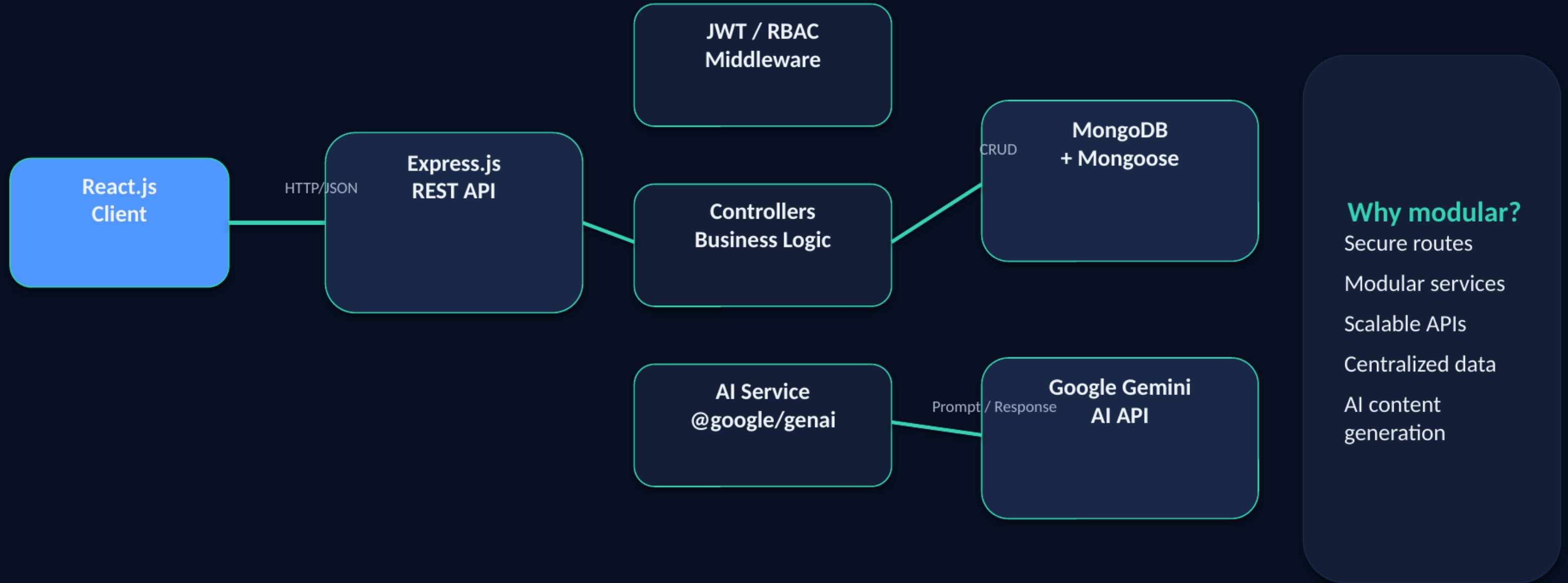
Long PDFs and notes are difficult to revise quickly.
Manual summaries, flashcards and questions consume study time.
Students lack personalized plans based on exam timelines.
Multiple apps reduce productivity and learning continuity.
No centralized place for generated learning resources.

Solution

Upload / enter study material through a secure backend.
Gemini AI generates summaries, flashcards and MCQ quizzes.
Personalized study plans use goals, available time and exam date.
MongoDB stores materials and generated resources.
JWT authentication + RBAC protects private and admin routes.

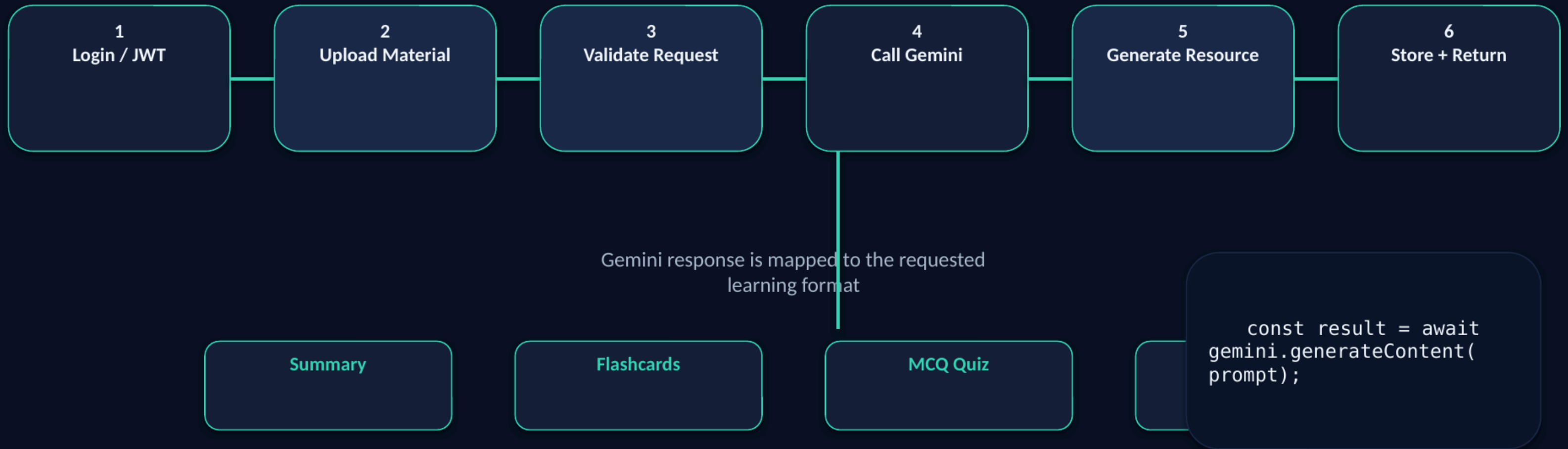
2. System Architecture

Modular RESTful architecture separates client, security, business logic, AI and data



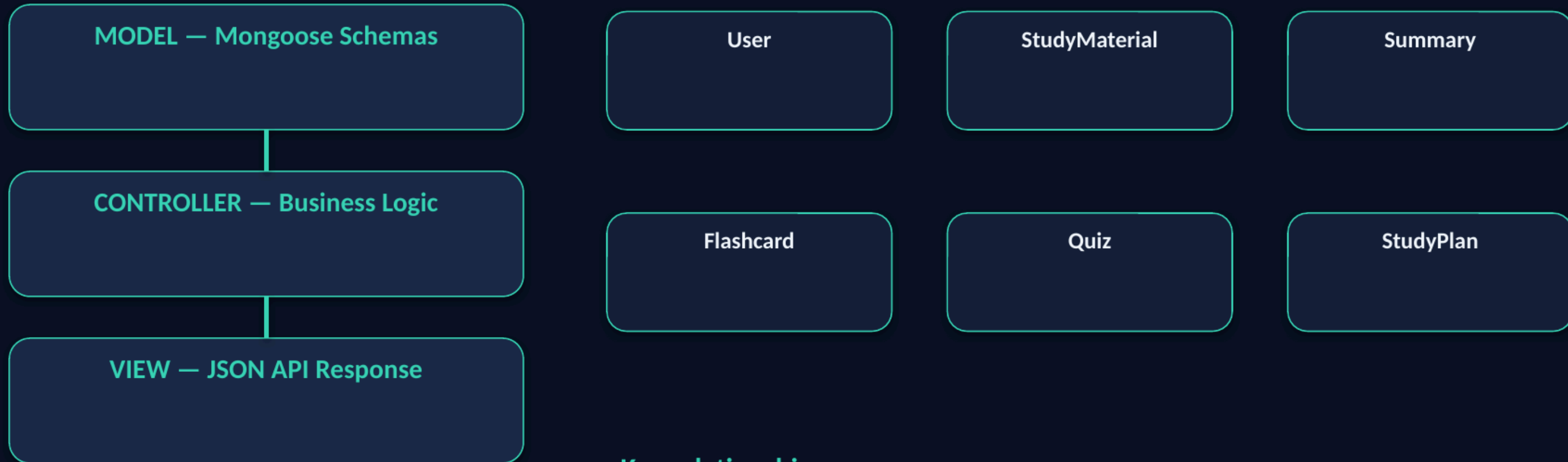
3. AI Content Generation Flow

Core runtime flow for summary / flashcards / quiz / study-plan generation



4. MVC + MongoDB Data Model

Separation of concerns keeps the backend maintainable and scalable

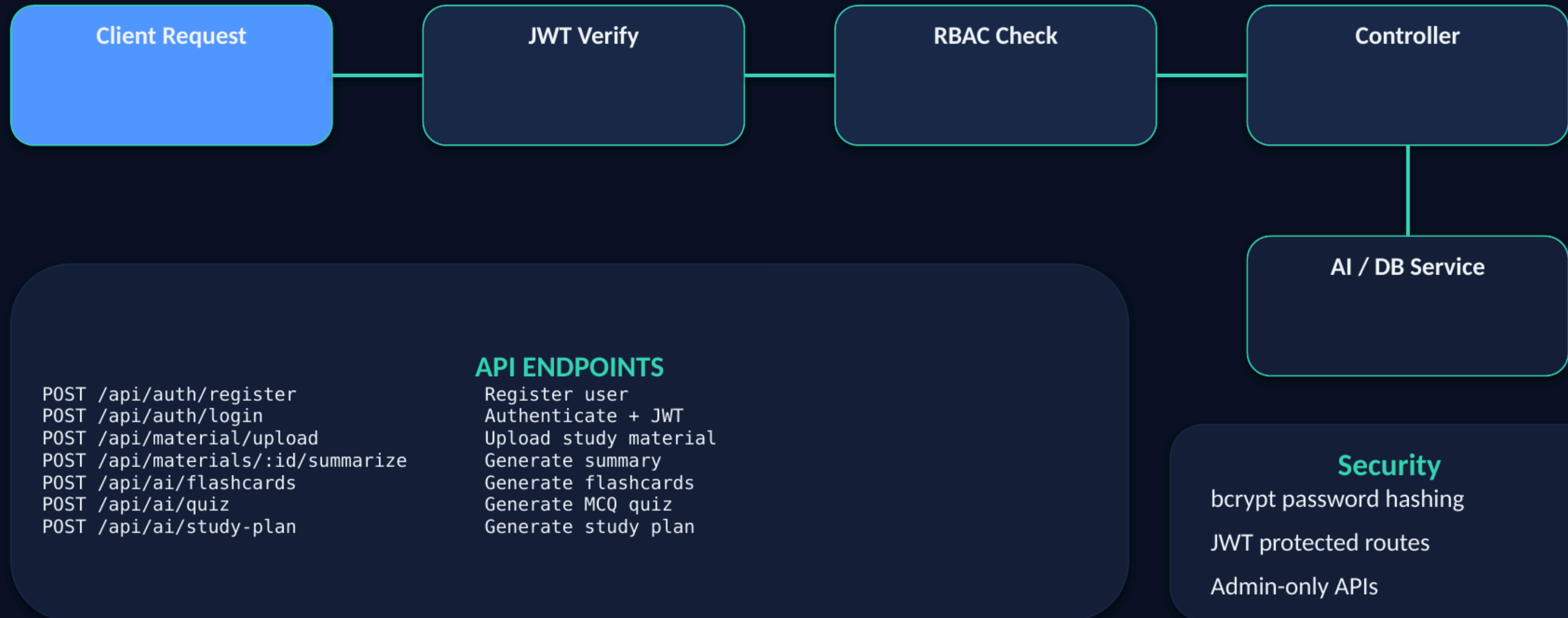


Key relationships

User → Study Material (1:N) | Study Material → Flashcards (1:N)
Study Material → Summary (1:1) | Study Material → Quiz (1:1)
User → Study Plan (1:N)

5. Security & REST API Layer

Authentication, authorization and API endpoints protect the learning workflow



6. Implementation & Outcome

A deployable foundation for personalized AI-assisted learning

Implementation

Node.js + Express.js backend with modular REST APIs.

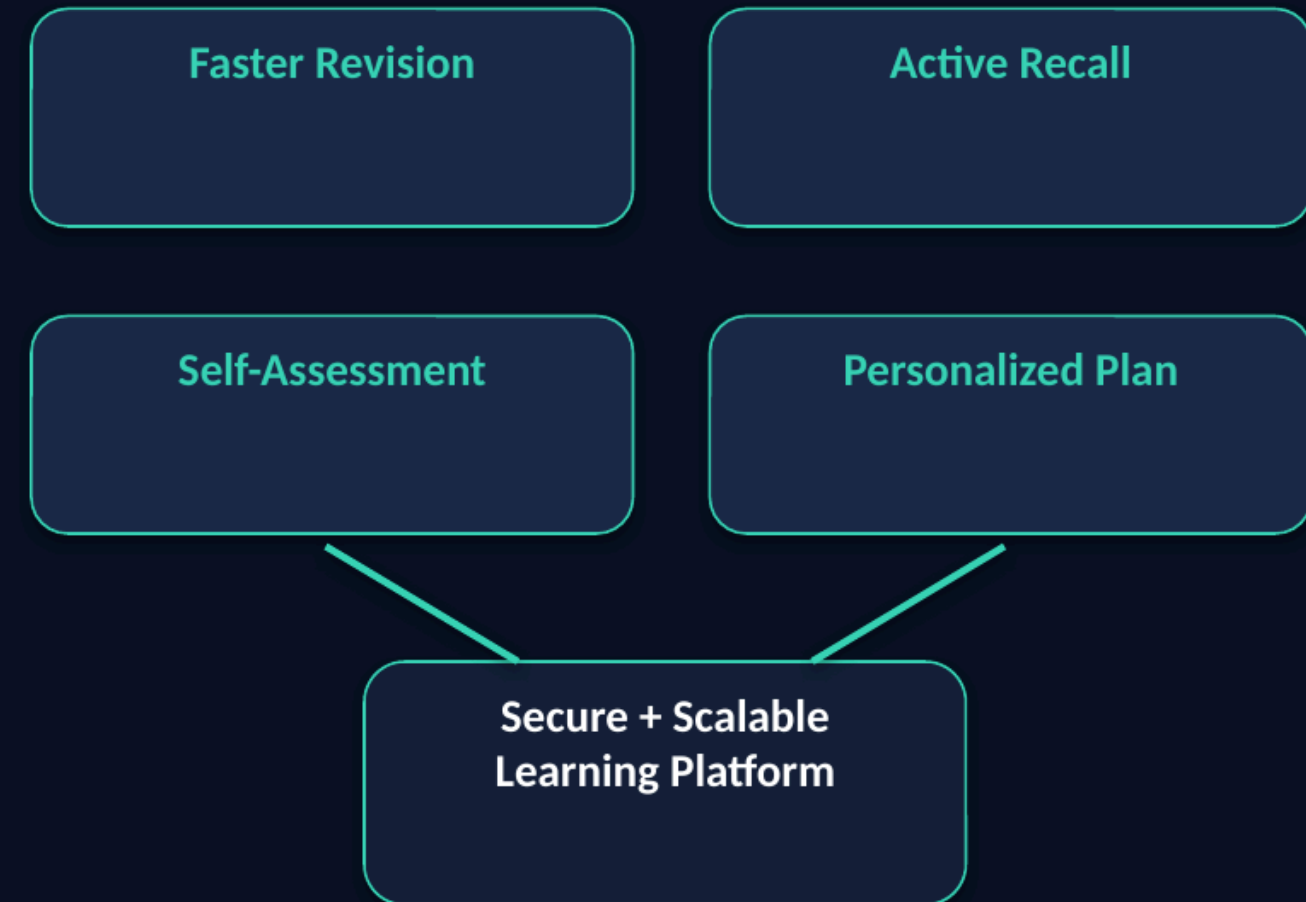
MongoDB + Mongoose for schema validation and data persistence.

Google Gemini API for dynamic educational content generation.

Multer-based upload validation for study materials.

Environment variables store PORT, MONGO_URI, JWT_SECRET and GEMINI_API_KEY.

Postman / Thunder Client can verify REST endpoints.



Outcome: reduced manual preparation and a centralized workflow for study resources.